

Phase-Shifting Transformers: Quadrature Injection Principle

How to rotate an AC voltage phasor?

- In balanced 3-phase, the line-to-line voltage between the **other two phases** is strictly orthogonal (90°) to phase a :

$$\bar{V}_b = V e^{-j120^\circ}, \quad \bar{V}_c = V e^{j120^\circ}$$

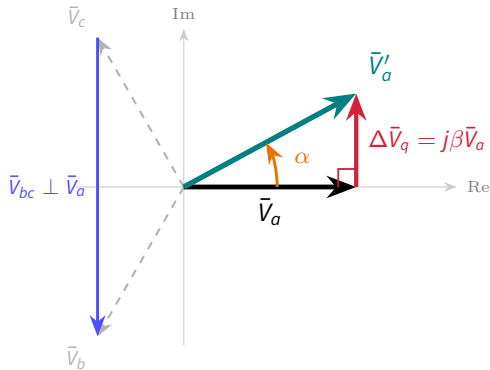
$$\bar{V}_{bc} = \bar{V}_b - \bar{V}_c = -j\sqrt{3}\bar{V}_a \quad (\perp \bar{V}_a!)$$

- By injecting a controllable fraction $\pm k$ of $\bar{V}_{bc}$ **in series** with phase a :

$$\bar{V}'_a = \bar{V}_a + \Delta\bar{V}_q = \bar{V}_a \pm jk\sqrt{3}\bar{V}_a$$

- The resulting phasor is rotated by an angle α :

$$\alpha = \arctan\left(\frac{|\Delta\bar{V}_q|}{|\bar{V}_a|}\right) \approx \pm k\sqrt{3}$$



Phase-Shifting Transformers: Two-Core Architecture & Operation

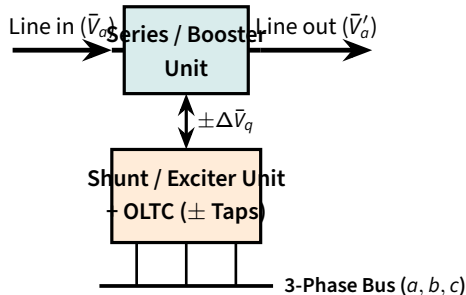
Substation Implementation (Two-Core Design):

- ▶ **1. Shunt / Exciter Unit:** Connected line-to-line across phases (bc, ca, ab) to tap quadrature voltages.
- ▶ **2. On-Load Tap Changer (OLTC):** Adjusts the turns ratio and flips reversing switch to provide discrete steps:

$$\alpha \in [-17, +17] \times 1.5^\circ \quad (\text{e.g. Elia 380 kV})$$

- ▶ **3. Series / Booster Unit:** Injects $\Delta \bar{V}_q$ directly in series into the transmission line.

Symmetrical vs. Asymmetrical:



Operational Function: The Electrical Valve

By modifying the voltage angle across a corridor without generator redispatch, PSTs: